

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458472.5977 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.022
Transit depth (Rp/Rs)^2	4.63 +/- 0.94 [%]
Orbital Inclination (inc)	83.84 +/- 1.65
Airmass coefficient 1 (a1)	1.065 +/- 0.014
Airmass coefficient 2 (a2)	-0.042 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#3 - [160, 28]
Optimal Aperture	3.94
Optimal Annulus	11.61
Transit Duration (day)	0.047 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.215 $\pm$ 0.022 vs 0.1646 (deviation 2.29 σ)
Duration fit vs published	0.047 d vs 0.0758 d (deviation 2.88 σ)
Mid-time O-C	1.5 min
Depth SNR	9.8
Residual scatter	1.36 %

Light curve

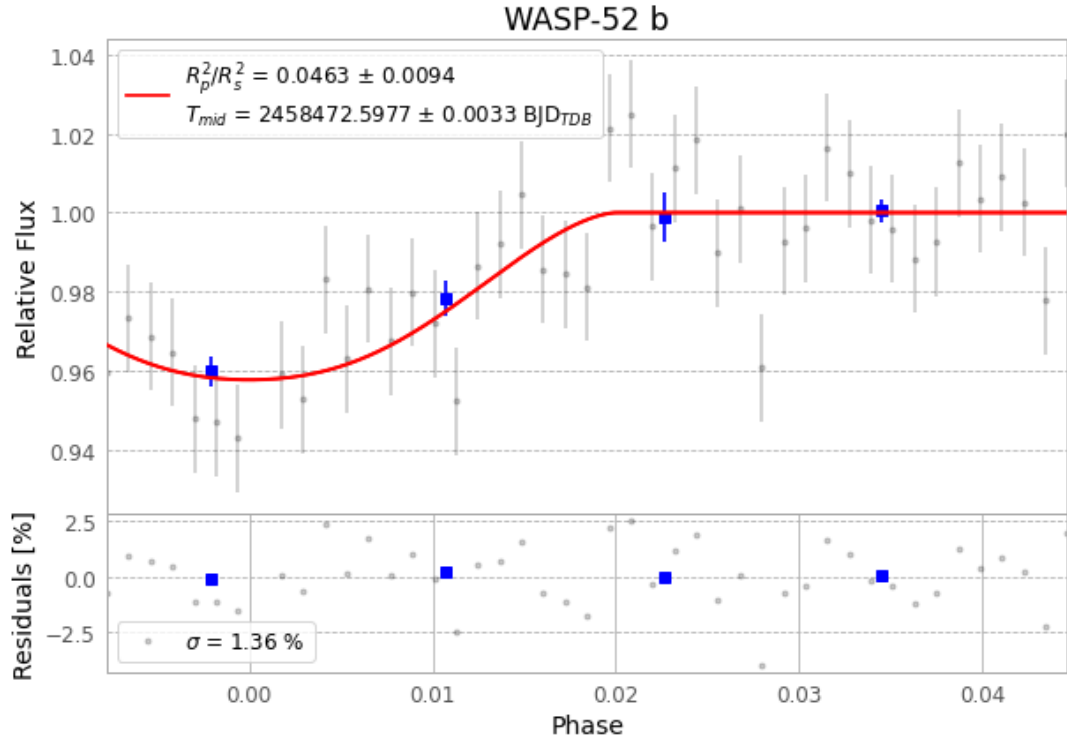


Figure 1 — FinalLightCurve_WASP-52 b_2018-12-19.png (SHA-256 6114cfc447cefd80...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [256, 319], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0f2d02aae654e512...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

45 FITS frames (30 MB), WASP-52181220020012.FITS → WASP-52181220041212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-181220.log (aa892fb96931...), FinalLightCurve_WASP-52 b_2018-12-19.png (6114cfc447ce...), FinalLightCurve_WASP-52 b_2018-12-19.csv (389514beb4ae...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 04:10Z.